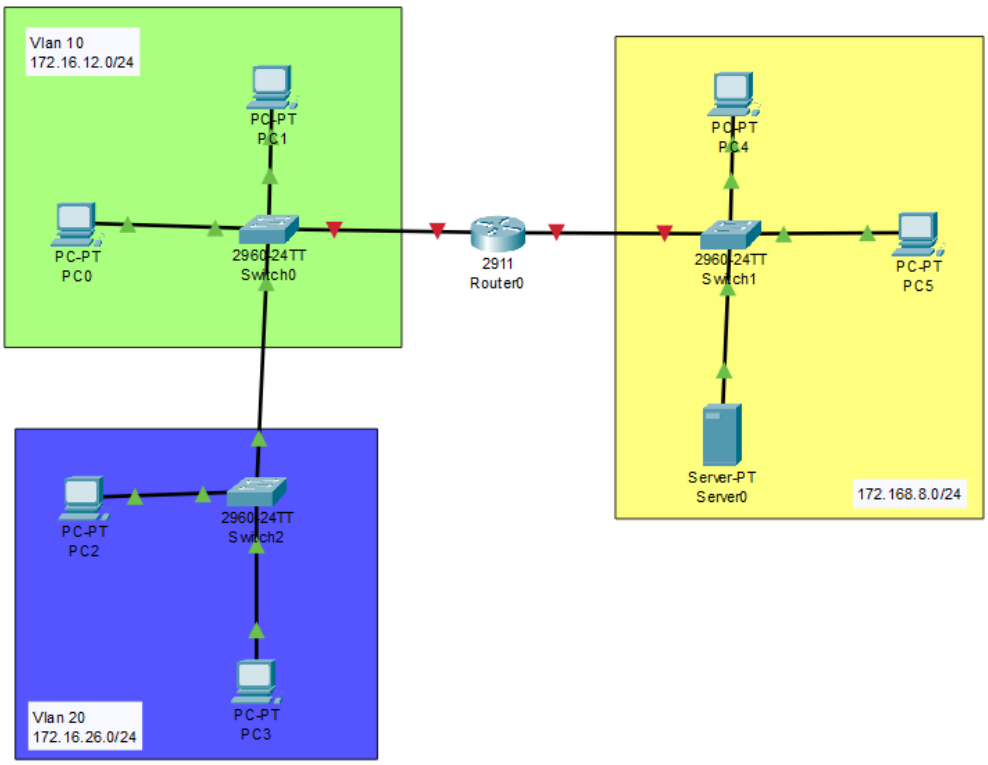
Практическа работа 3

1) Настройкак топологии согласно схеме(M = 6, N = 2):



2) Настроим маршрутизатор и его порты:

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interface gigabitEthernet 0/0
R1(config-if)#exit
R1(config)#interface gigabitEthernet 0/1
R1(config-if)#ip address 192.168.8.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R1(config-if)#exit
R1(config)#interface gigabitEthernet 0/0.10
R1(config-subif)#encapsulation dot1Q 10
R1(config-subif)#ip address 172.16.12.1 255.255.255.0
R1(config-subif)#ip helper-address 192.168.8.2
R1(config-subif)#exit
R1(config)#interface gigabitEthernet 0/0.20
R1(config-subif)#encapsulation dot1Q 20
R1(config-subif)#ip address 172.16.26.1 255.255.255.0
R1(config-subif)#ip helper-address 192.168.8.2
R1(config-subif)#exit
R1(config)#interface gigabitEthernet 0/0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
```

3) Настроим Switch и его порты:

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#vlan 20
Switch(config-vlan)#exit
Switch(config)#interface giga
Switch(config)#interface gigabitEthernet 0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

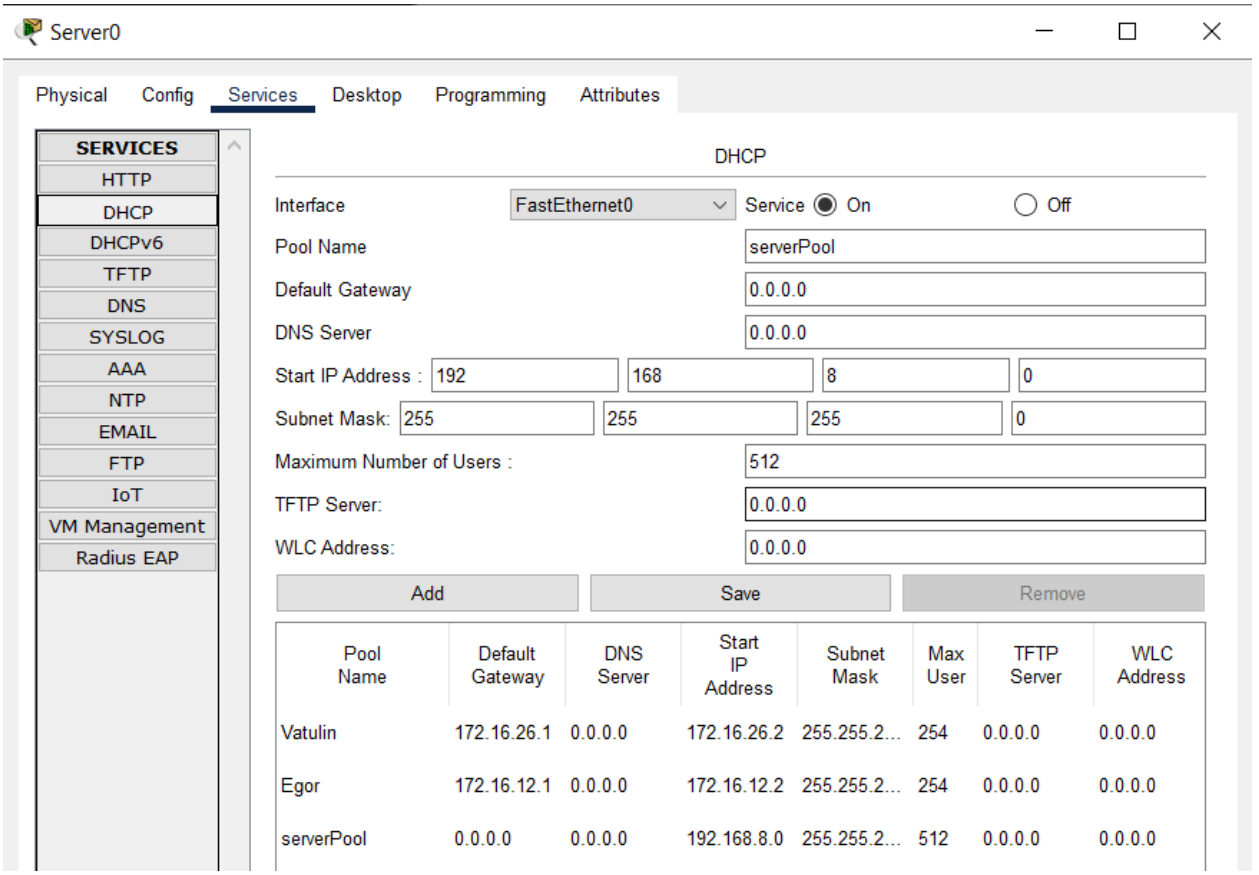
Switch(config-if)#exit
Switch(config)#interface gigabitEthernet 0/2
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

Switch(config-if)#exit
Switch(config)#interface range fa0/1-2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
```

4) Настройка DHCP сервера:

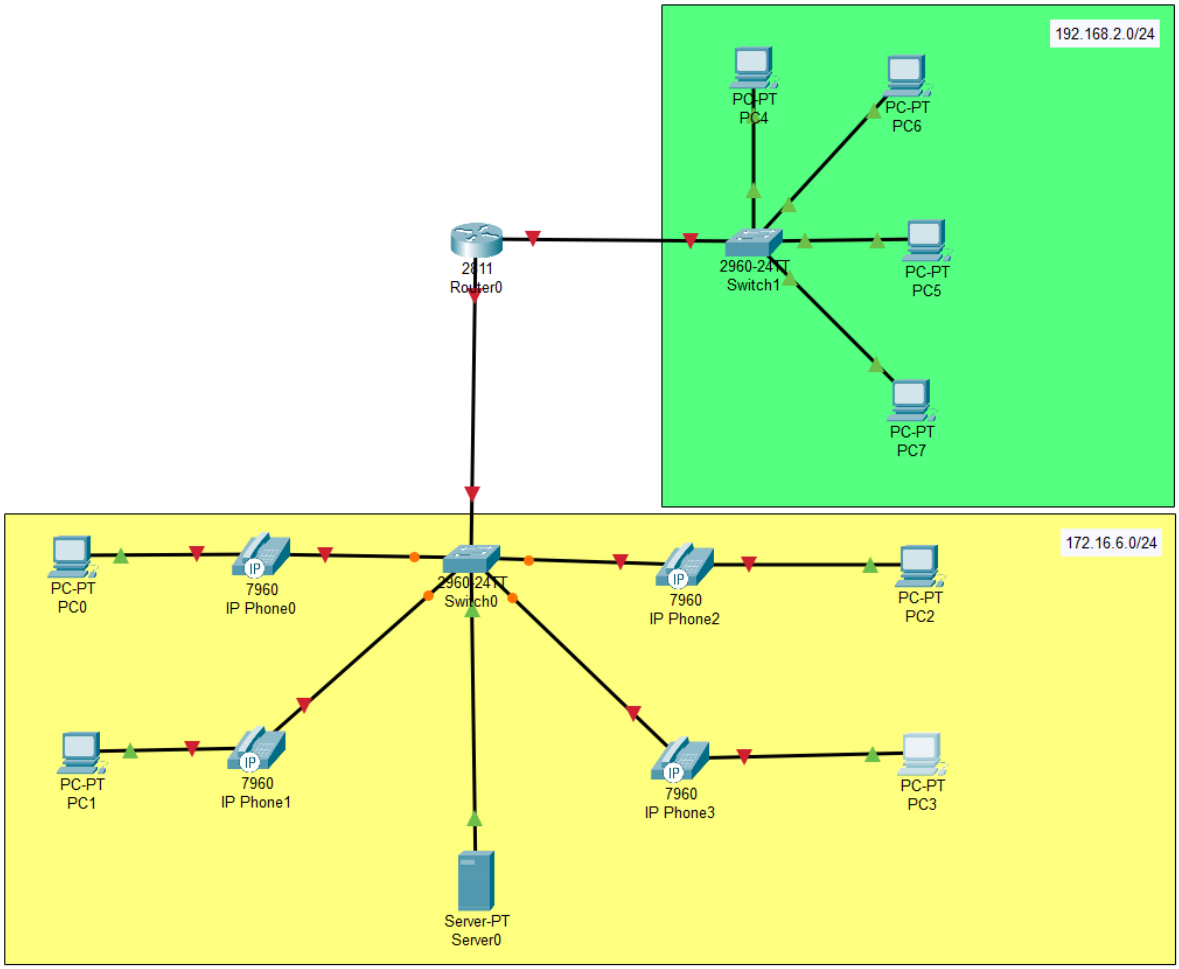


5) Команда show ip dhcp binding:

```
R1>show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
```

Практическа работа 4

1) Настроим топологию согласно рисунку



2) Назначим имя маршрутизатору и настроим Fa0/1 и Fa0/0:

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interface fa0/0
R1(config-if)#ipaddress 192.168.2.1
                 ^
% Invalid input detected at '^' marker.

R1(config-if)#ip address 192.168.2.1
% Incomplete command.
R1(config-if)#ip address 192.168.2.1
% Incomplete command.
R1(config-if)#ip address 192.168.2.1 255.255.255.0
R1(config-if)#exit
R1(config)#interface fa0/1
R1(config-if)#ip address 172.16.6.1 255.255.255.0
R1(config-if)#
```

Настроим на маршрутизаторе dhcp-poll и option:

```
R1>en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip dhcp pool voice
R1(dhcp-config)#network 172.16.6.0 255.255.255.0
R1(dhcp-config)#default-router 172.16.6.1
R1(dhcp-config)#ipdhcpexcluded-address 172.16.6.1 172.16.6.3
                 ^
% Invalid input detected at '^' marker.

R1(dhcp-config)#ip dhcp excluded-address 172.16.6.1 172.16.6.3
R1(config)#ip dhcp pool voice
R1(dhcp-config)#option 150 ip 172.16.6.1
```

Настраиваем VoIP параметры:

```
R1>en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#telephony-service
R1(config-telephony)#max-dn 4
R1(config-telephony)#max-ephones 4
R1(config-telephony)#ip source-address 172.16.6.1 port 3100
R1(config-telephony)#auto assign 1 to 19
```

3) Настроим номера для телефонов:

```
R1(config)#ephone-dn 1
R1(config-ephone-dn)#%LINK-3-UPDOWN: Interface ephone_dsp DN 1.1, changed state to up

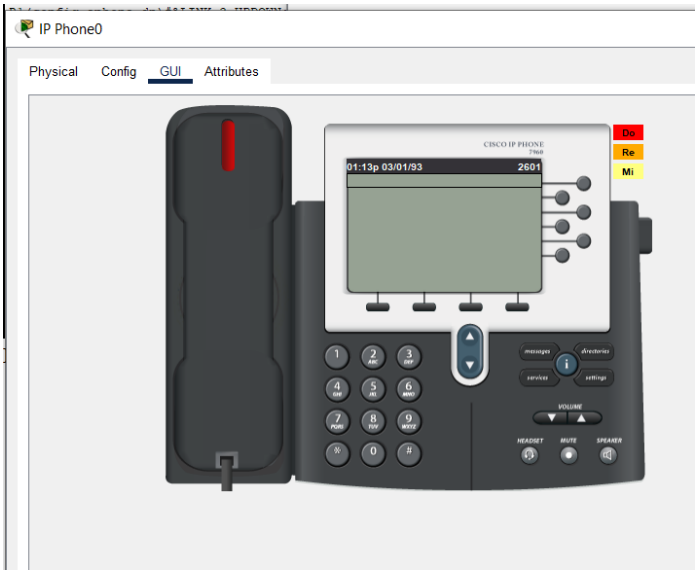
R1(config-ephone-dn)#number 2601
R1(config-ephone-dn)#exit
R1(config)#ephone-dn 2
R1(config-ephone-dn)#%LINK-3-UPDOWN: Interface ephone_dsp DN 2.1, changed state to up

R1(config-ephone-dn)#number 2602
R1(config-ephone-dn)#exit
R1(config)#ephone-dn 3
R1(config-ephone-dn)#%LINK-3-UPDOWN: Interface ephone_dsp DN 3.1, changed state to up

R1(config-ephone-dn)#number 2603
R1(config-ephone-dn)#exit
R1(config)#ephone-dn 4
R1(config-ephone-dn)#%LINK-3-UPDOWN: Interface ephone_dsp DN 4.1, changed state to up

R1(config-ephone-dn)#number 2604
R1(config-ephone-dn)#exit
R1(config)#
%IPPHONE-6-REGISTER: ephone-1 IP:172.16.6.4 Socket:2 DeviceType:Phone has registered.
```

4) Получим номер для каждого телефона:









5) Настроим Server:

DHCP

Interface FastEthernet0 Service On Off

Pool Name Vatulin

Default Gateway 172.16.6.1

DNS Server 0.0.0.0

Start IP Address : 192 168 2 4

Subnet Mask: 255 255 255 0

Maximum Number of Users : 5

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

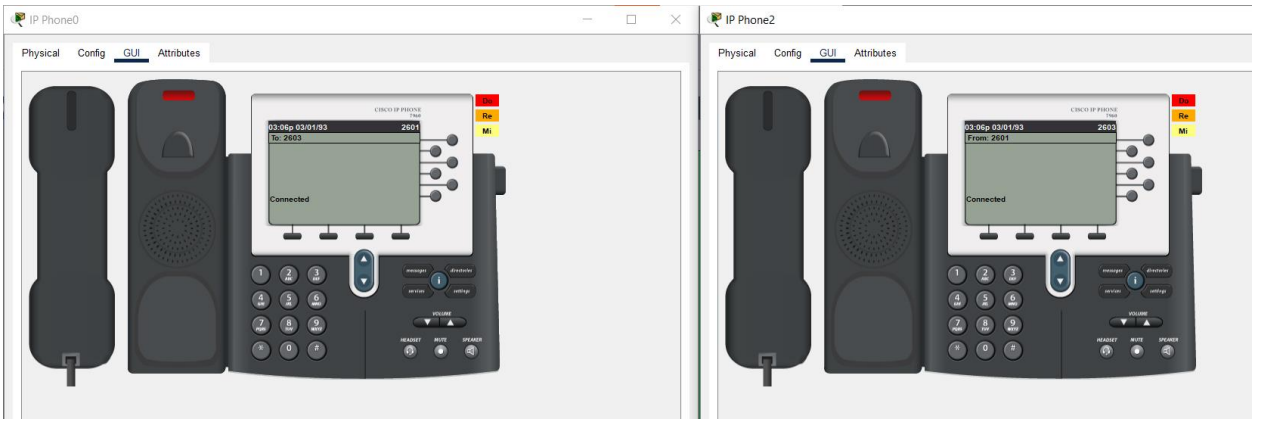
Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
Vatulin	172.16.6.1	0.0.0.0	192.168.2.4	255.255.2...	5	0.0.0.0	0.0.0.0

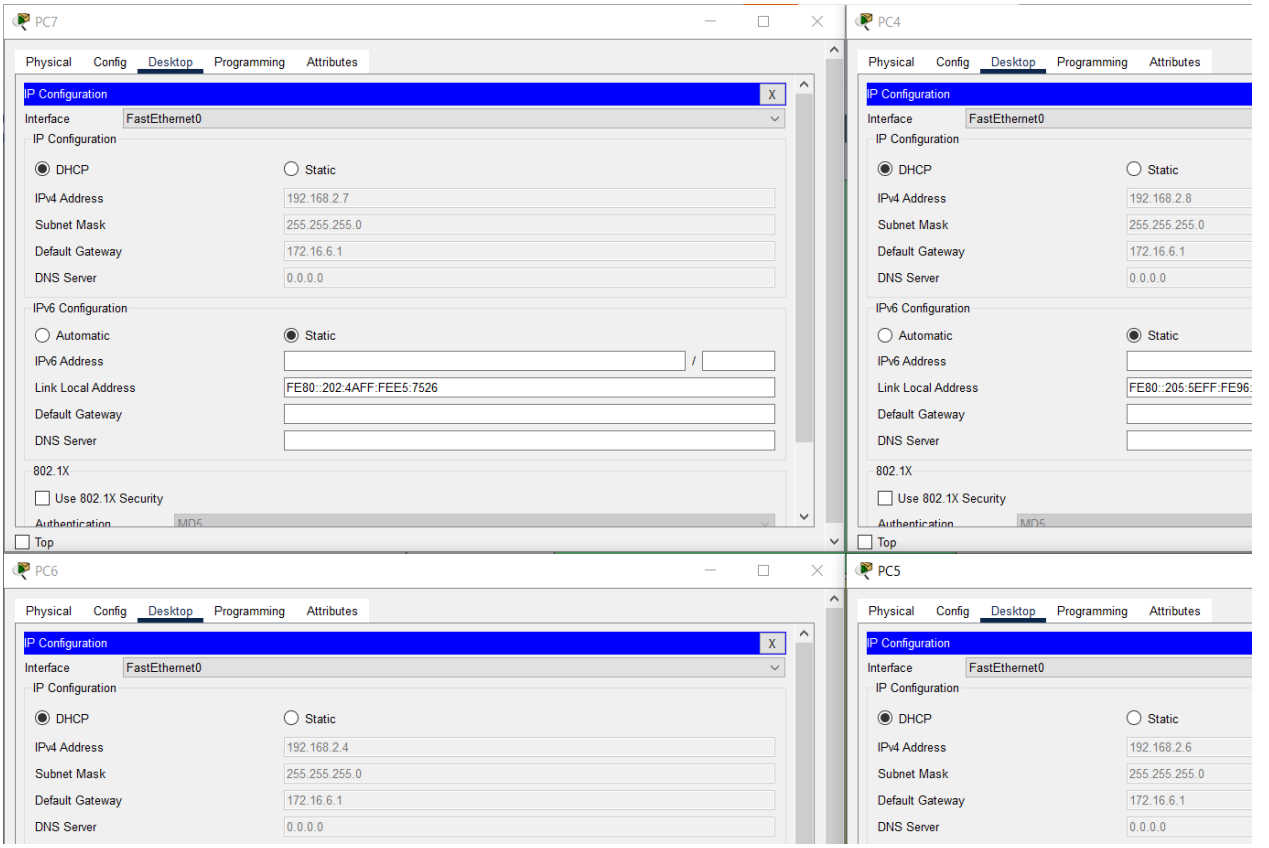
6) Команда show ip dhcpb inding:

```
R1#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
172.16.6.5       0002.4A23.B7DB            --                    Automatic
172.16.6.7       0030.F23B.1E3E            --                    Automatic
172.16.6.9       0030.F23E.354B            --                    Automatic
172.16.6.10      000A.4100.6C5E            --                    Automatic
172.16.6.4       0010.11A6.7723            --                    Automatic
172.16.6.13      0001.C76B.0331            --                    Automatic
```

7) Звонок между телефонами:



8) Адреса на компьютерах 4-7:



9) show mac-address-table на Switch1:

```
Switch#show mac-address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0002.4ae5.7526    DYNAMIC     Fa0/4
   1    0005.5e96.5572    DYNAMIC     Fa0/1
   1    000a.4102.ca01    DYNAMIC     Gig0/1
   1    00d0.d39c.3e30    DYNAMIC     Fa0/3
   1    00e0.a30e.bccd    DYNAMIC     Fa0/2
```

10) show run на Router:

```
R1>show run
         ^
% Invalid input detected at '^' marker.

R1>en
R1#show run
Building configuration...

Current configuration : 1452 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R1
!
!
!
!
ip dhcp excluded-address 172.16.6.1 172.16.6.3
!
ip dhcp pool voice
 network 172.16.6.0 255.255.255.0
 default-router 172.16.6.1
 option 150 ip 172.16.6.1
```